Class 19: Cancer mutation project

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I downloaded my sequences and moved them to my working project directory.

I ran a BLASTp search against REFSEQ_PROTEIN filtered to human only. JOBID: VFJ13ST2016

Official Symbol: **BRAF** Official Full Name: **serine/threonine-protein kinase B-raf**

> Q2. What are the tumor specific mutations in this particular case (e.g. A130V)?

```
library(bio3d)

# Read my sequences
a <- read.fasta("A18116120_mutant_seq.fa")
```

We cab score conservation per sequnce position

```
s <- conserv(a)
mutation.position <- which( s !=1)
mutation.position
```

```
[1] 548 605 613 618
```

```
a$ali[1,mutation.position]
```

```
[1] "F" "S" "L" "L"
```

```
mutation.position
```

```
[1] 548 605 613 618
```

```r
a$ali[2, mutation.position]
```

```
[1] "V" "E" "R" "Y"
```

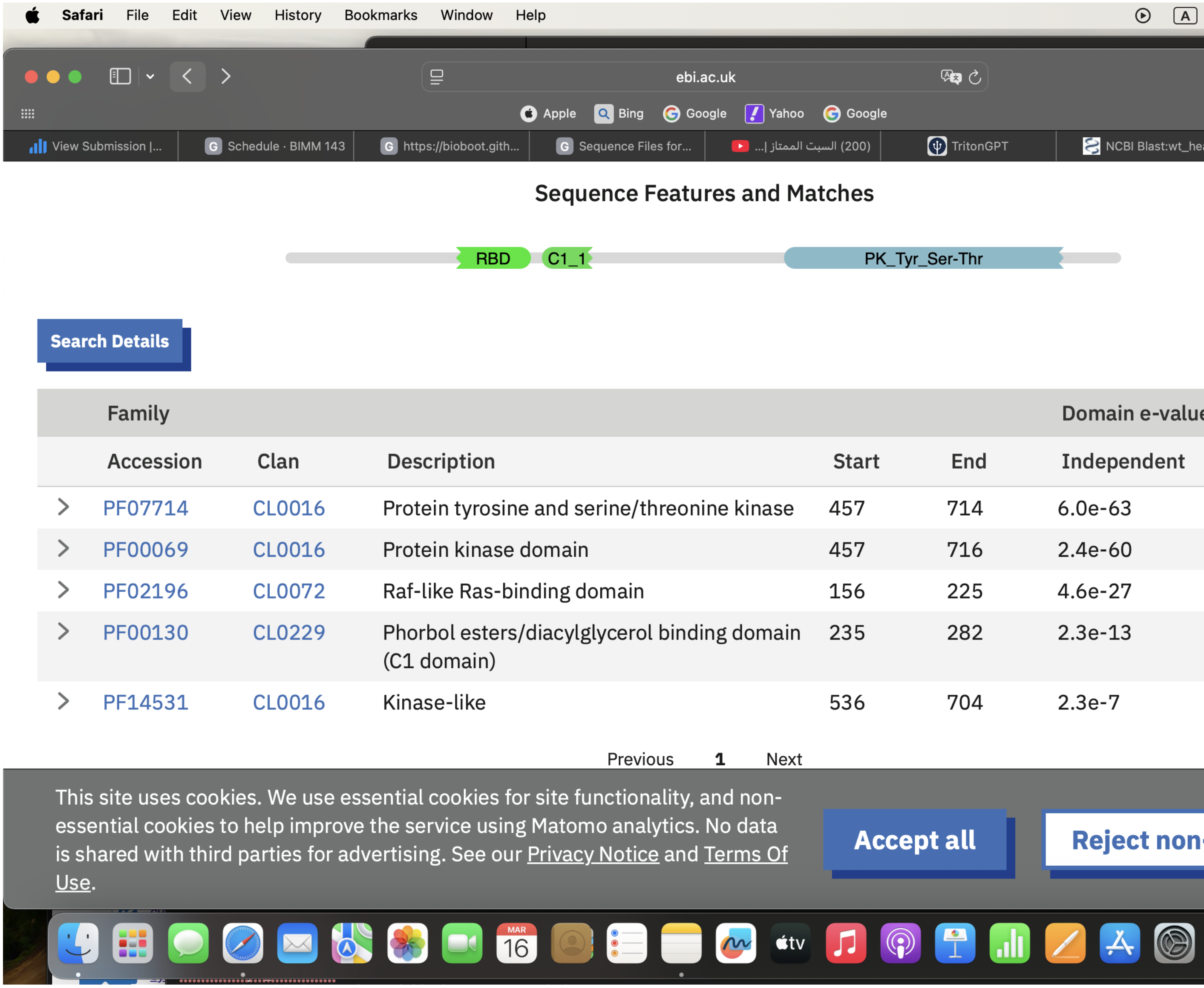
Answer: it seems for the code above that it will be F548V, S605E, L613R, L618Y

What PFAM domain do these mutation reside in?

We can use HMMER at the EBI (it works) to find this information

Answer: C-terminal region (residues ~457–716) corresponds to the protein kinase domain (F548V, S605E, L613R, L618Yfall within these domains). Not a single domian, but multi-domain protein. PFAM ID: PF07714 Domain name: Protein tyrosine and serine/threonine kinase (Pkinase)

```r
knitr::include_graphics("now.png")
```



Family					Domain e-value
Accession	Clan	Description	Start	End	Independent
PF07714	CL0016	Protein tyrosine and serine/threonine kinase	457	714	6.0e-63
PF00069	CL0016	Protein kinase domain	457	716	2.4e-60
PF02196	CL0072	Raf-like Ras-binding domain	156	225	4.6e-27
PF00130	CL0229	Phorbol esters/diacylglycerol binding domain (C1 domain)	235	282	2.3e-13
PF14531	CL0016	Kinase-like	536	704	2.3e-7

Q4. Using the NCI-GDC list the observed top 2 missense mutations in this protein (amino acid substitutions)?

Top 2 most common missense mutations in BRAF (NCI-GDC):

V640E and V640M

Q5. What two TCGA projects have the most cases affected by mutations of this gene? (Give the TCGA "code" and "Project Name" for example "TCGA-BRCA" and "Breast Invasive Carcinoma").

Answer: TCGA-THCA — Thyroid Carcinoma AND TCGA-SKCM — Skin Cutaneous Melanoma

> Q6.List one RCSB PDB identifier with 100% identity to the wt_healthy sequence and detail the percent coverage of your query sequence for this known structure? Alternately,provide the most similar in sequence PDB structure along with it's percent identity, coverage and E-value. Does this structure "cover" (i.e. include or span the amino acid residue positions) of your previously identified tumor specific mutations?

Answer:

Chain A, Serine/threonine-protein kinase B-raf [Homo sapiens]

PDB ID: 6QOK, chain A Percent identity: 99.87% Query coverage: 100% E-value: 0.0

It was found that this structure corresponds to the BRAF kinase domain (residues ~457–717) and has all F548V, S605E, L613R, L618Y (does cover all of the mutation positions.)